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ORIGINAL ARTICLE

Hematopoietic stem-cell transplantation in a zoo of multidrug-resistant organisms: Data from a cancer center in eastern India

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Abstract

Background: Infections by multidrug-resistant organisms (MDRO) are a major hurdle in hematopoietic stem-cell transplants (HSCTs). Conditioning regimens lead to mucosal barrier injury, which in-turn leads to transmigration of gut bacteria and sepsis. Pre-transplant stool and throat surveillance cultures can guide empirical antibiotic policy during the neutropenic period. In this paper, we document colonization with MDRO in pre-transplant surveillance cultures and the correlation with bloodstream infections in HSCT patients and analyze transplant outcomes with respect to these infections.

Methods: A single-center, retrospective study on HSCT was performed between January 2021 and December 2021. The incidence of bacterial infections, percentage of MDROs, correlation with pre-transplant stool/throat surveillance cultures, and their impact on overall 100-day and post-100-day to 6-month post-transplant survival were analyzed.

Results: Sixty-four patients were included in the study. Pre-transplant stool surveillance cultures were positive for MDRO in 85.9% of patients. Almost half (48.5%) of the isolates were positive for carbapenemase-producing genes (mainly New Delhi metallo-beta-lactamase-1 [NDM-1] and oxacillinase-48 [OXA-48]). Eighteen patients (18/64, 28%) had a positive blood culture for MDRO in the peri-engraftment neutropenic period. Correlation between surveillance and blood cultures was seen in 61% (11/18) of patients. All-cause mortality was 14.1% (9/64) and 25% (16/64) in patients at 100 days and 6 months post-HSCT, respectively. The 100-day and post-100-day all-cause mortality rates were higher in patients with Gram-negative MDRO bloodstream infections ($p < .012$ and $< .008$, respectively).

Conclusion: MDRO infections can adversely affect HSCT outcomes. Pre-transplant stool and throat surveillance cultures may guide empirical antibiotic policy and lead to favorable transplant outcomes.



KEYWORDS

Gram-negative sepsis, HSCT, MDRO, surveillance culture, transplant outcomes

1 | INTRODUCTION

Bacterial infections due to multidrug-resistant organisms (MDRO) are a major hurdle in the setting of hematopoietic stem-cell transplants (HSCT). MDRO are defined as organisms, predominantly bacteria, resistant to at least one agent in three or more classes of antimicrobials.¹ Mucosal barrier injury leading to transmigration of pathogenic organisms from the gut into the bloodstream, known by the term mucosal barrier injury-laboratory confirmed bloodstream infections (MBI-LCBI) is a leading cause of post-transplant sepsis.¹ Transmigration of MDRO can lead to increased morbidity and mortality due to neutropenic sepsis, increased risk of graft versus host disease (GVHD), healthcare costs, and healthcare resource utilization.² It is therefore prudent to screen patients pre-transplant, for colonization of the gut by resistant organisms and study their genotypes, resistance mechanisms, and antibiotic susceptibility profile. This enables not only barrier precautions but also, a more informed selection of appropriate empirical antibiotics.^{3–5} Ours is a tertiary-care cancer center in eastern India performing about 50 autologous and allogeneic transplants in a year. The bone marrow transplant unit is High Efficiency Particle Arrester filtered and positive differential pressure of 2.5 Pascal higher than the ante-room/corridor is maintained. Patient rooms are fitted with Pall 0.2- μ m water filters in the shower heads and faucets.^{6,7} Regular audits for hand hygiene compliance, air and water quality, and sterilization/disinfection are done to ensure that standards are met. Despite these strict measures, our center has seen a considerable number of MDRO bloodstream infections in recent years. The same trends have also been identified in other areas, such as hematology wards and intensive care units, with the possible source of infection being the patient's own gastrointestinal tract. To ascertain the impact of MDRO infections on HSCT outcomes, we analyzed the data of our transplant unit for MDRO prevalence rates, the incidence of Gram-negative bacterial bloodstream infections, and correlations, if any, between pre-transplant surveillance cultures and subsequent bloodstream infections in these patients. We also analyzed the trends of MDRO prevalence and antibiotic resistance patterns over the preceding 7 years.

2 | METHODS

2.1 | Study design and parameters evaluated

This was a single-center retrospective observational study of all HSCTs performed at our center over one calendar year (1 January 2021–31 December 2021) to evaluate the incidence of bacterial bloodstream infections (BSIs), the MDRO positivity rates in BSIs, MDRO coloniza-

tion rates, correlation of pre-transplant stool and throat surveillance cultures with BSIs, and the overall impact of Gram-negative BSI on all-cause 100-day and 100 days–6 months post-HSCT mortality. As per our departmental policy, stool and throat surveillance cultures are sent for all HSCT patients, initially, 1–2 weeks prior to admission to the transplant unit, and then weekly until neutrophil engraftment occurs. In this study, stool and throat surveillance was done at a median of 10 days before transplant (interquartile range: 6–20 days for stool and 7–20 days for throat swabs). The clinical infectious disease team of our hospital is instrumental in antibiotic stewardship and establishes, implements, and updates antibiotic policy regularly. The Infectious Diseases Society of America's guideline for the use of antimicrobial agents in neutropenic patients with cancer guides our empiric antibiotic policy for febrile neutropenia in patients undergoing leukemia induction and HSCT.⁸ In addition, antibiotic policy (empiric and escalation plan) for each patient undergoing an autologous or allogeneic stem-cell transplant (SCT) is tailor-made prior to admission, based on the stool and throat surveillance cultures.

2.2 | Definitions

Multidrug-resistant (MDR) bacteria: Resistant to at least one agent in three or more classes of antibiotics.

Extensively drug-resistant (XDR) bacteria: Bacterial isolates susceptible to only one or two of all available antimicrobial categories.^{1,9}

Empirical antimicrobial therapy: Antimicrobial therapy among patients with suspected neutropenic fever or sepsis consisted of the following regimens: first-line—piperacillin-tazobactam with amikacin; second line—meropenem with teicoplanin; third line—meropenem with colistin/polymyxin B.

Directed antimicrobial therapy: Teicoplanin was added for Methicillin Resistant *Staphylococcus Aureus* (MRSA), daptomycin for vancomycin-resistant *Enterococcus* (VRE), and meropenem for extended spectrum beta-lactamase (ESBL) and Ambler class C-producing beta-lactamases (AmpC). For carbapenem-resistant organisms (CRO), we used meropenem + colistin/polymyxin B. For carbapenemase-producing *Klebsiella pneumoniae* (KPC) or OXA-48-positive CRO, we used ceftazidime-avibactam with/without colistin/polymyxin B. For New Delhi metallo-beta-lactamase (NDM)-positive CRO, we used ceftazidime-avibactam with aztreonam.¹⁰ Tigecycline and fosfomycin were added to the third-line agents in cases of skin/soft tissue/intra-abdominal infection or urinary tract infection, respectively. Antimicrobial therapy was tailored based on results of surveillance cultures, blood, and other infection site cultures, response to therapy (clinical/microbiological), and adjusted based on renal/liver function. In febrile patients who were tachypneic

or hypotensive, antibiotics were escalated to the highest level (ceftazidime-avibactam $\pm$ aztreonam) based on the surveillance culture and sensitivity without going through the empiric algorithm.

3 | MICROBIOLOGICAL METHODS

3.1 | Surveillance culture for MDR/XDR bacteria from stool and throat swab samples

Surveillance culture for VRE, third-generation cephalosporin-resistant (3GCR) *Enterobacterales* (ESBL/AmpC), and CRO were performed based on a modification of the method described by Landman et al.¹¹ The method had been previously standardized, validated and used in published studies.¹² Genotyping of the CRO was done by an endpoint polymerase chain reaction (PCR)-based assay previously validated.¹³ For stool samples, 4 mL of sterile saline and a loop of stool (0.5 g) were taken in a test tube and made into a uniform suspension. Four test tubes, each containing 4 mL of trypticase soya broth (TSB), were then set up, and 100 μ L of stool suspension was added to each tube. Antibiotic susceptibility testing discs of cefpodoxime (30 μ g), ceftazidime (30 μ g), meropenem (10 μ g), and meropenem (10 μ g) with vancomycin (30 μ g) were then added. One sterile saline tube acted as a negative control. All these tubes were incubated at 37°C overnight (16–18 h). Subculture was performed from all the tubes on MacConkey agar plates. The plates were incubated overnight and assessed for colonies presumptively resistant to respective antibiotics. VRE is detected by selective enrichment culture using vancomycin and meropenem discs in a broth. As VRE isolates are usually *Enterococcus faecium* (which are ampicillin and meropenem resistant), they are selected out by this enrichment culture. The identity of the organism is verified biochemically by standard microbiological tests, and van A gene, which is present in about 95% of the VRE isolates in our setting, is checked with a real-time PCR. ESBL producers were distinguished from the AmpC producers using double disc test. In case of ESBL, ceftazidime and ceftazidime + clavulanic acid were used. In case of AmpC detection, ceftazidime and ceftazidime + cloxacillin were used. An enhancement of zone diameters by greater than or equal to 5 mm was confirmatory for ESBL and AmpC production, respectively. Identity of the organism was confirmed biochemically using standard microbiological tests.

For throat swab samples, HiMedia swab MS651, sterile cotton tipped swabs in Amies medium and charcoal in polypropylene tube was used. The swab was dipped into 4 mL of TSB, which was then incubated overnight at 37°C. The subsequent process of identification of resistant colonies was similar to that used for stool samples.

3.2 | Blood cultures and antibiotic susceptibility testing

Bloodstream infection (bacteremia or fungemia) was detected using BACT/ALERT system (Biomérieux, USA). Antimicrobial susceptibility

for bacteria and fungi (Candida/yeasts) was performed using VITEK-2 Compact System (Biomérieux, USA) (Figure 1).

3.3 | Quality control

Quality control of both VITEK- and disc-diffusion-based tests was done using ATCC strains of *Escherichia coli* (ATCC25922; ATCC35218—ESBL), *K. pneumoniae* (ATCC700603—ESBL), *Pseudomonas aeruginosa* (ATCC27853), *Staphylococcus aureus* (ATCC29213; BAA-976; BAA-977—for inducible clindamycin resistance; ATCC93300—MRSA; ATCC25923—for manual antimicrobial susceptibility testing), and *Enterococcus faecalis* (ATCC51299). The laboratory also participated in the external quality assurance program of the Indian Association of Medical Microbiology (Figure 1).

3.4 | Statistical methods

Patients' demographic data, transplant types, and microbiological data were represented as descriptive statistics in mean and median with standard deviation, while predictors of mortality (categorical variables) were analyzed using chi-square test and Fisher's exact test. All analyses were done using IBM SPSS version 23 (IBM Inc., Armonk, USA). All analyses were two-tailed, and level of significance was kept at $<.05$.

4 | RESULTS

Between January 2021 and December 2021, our center performed 64 HSCTs (29 autologous and 35 allogeneic transplants—12 matched related donors, five matched unrelated donors, and 18 haploidentical transplants). The median age of the patient population was 37.5 years (range 2–67 years). Male patients accounted for 65.6% (42/64) of the cohort. Autologous transplants were done for Hodgkin and non-Hodgkin lymphomas (HL and NHL), multiple myelomas, and solid tumors such as neuroblastomas and Ewing's sarcoma. Allogeneic transplants were done for various indications, such as high-risk/relapsed/refractory acute leukemias in remission, beta thalassemia major, myelofibrosis, refractory T-cell lymphomas, and refractory HL and NHL, which relapsed after autologous transplants. All patients had good performance status (Eastern Cooperative Oncology Group¹⁴ performance status scale—1 [$n = 58$], 0 [$n = 6$]) and were free of active infection prior to transplant, and all were tested negative for SARS CoV-2 by RT-PCR prior to admission in the transplant unit (Table 1).

Stool surveillance cultures yielded significantly more MDRO than throat swab surveillance cultures (55/64 [85.9%] vs. 21/64 [32.8%]) (Table 1). *E. coli* was the most common MDRO in stool samples (AmpC 19/64 [29.6%], ESBL 22/64 [34.3%], and carbapenemase producers 16/64 [25%]), whereas *Klebsiella* was more common MDRO in throat swab surveillance cultures. Twenty out of 64 (31.25%) throat swab surveillance cultures were found to be positive for MDRO, of which

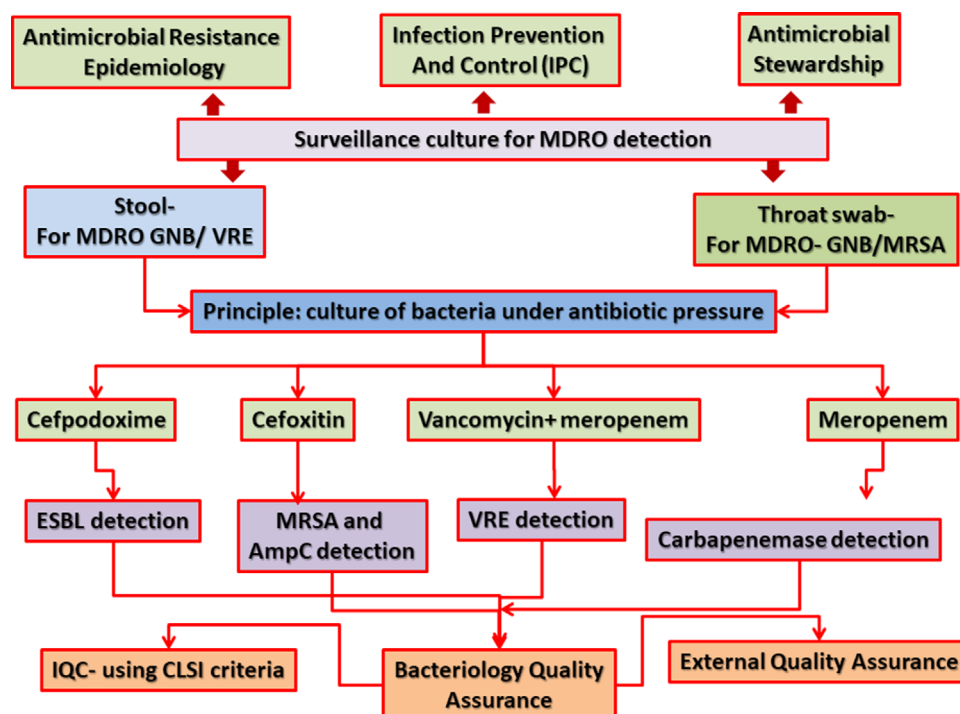


FIGURE 1 Graphical representation of the microbiological methods and quality control strategies used at our institution for the detection of multidrug-resistant organism (MDRO) by stool and throat surveillance cultures.

11 (17.18%) were carbapenem resistant. Carbapenemase-producing *Enterobacterales* (CPE) with NDM and/or OXA-48 genes (CPE—*E. coli* and *Klebsiella*) were found in 48% (31/64) and 9.3% (6/64) of all the stool and throat isolates, respectively. Overall carbapenem resistance in surveillance cultures was seen in 48.5% (31/64) of patients (Tables 1 and 2).

A total of 18 Gram-negative BSI events were documented in 64 transplants (28.1%, 18/64). Most common isolate was MDRO *Klebsiella* species (9/18, 50%) followed by *E. coli* (8/18, 44.4%). *P. aeruginosa* isolates accounted for 3/18 isolates (Table 2). Six patients had BSI with multiple Gram-negative bacteria isolated viz. *E. coli* + *Klebsiella*, *E. coli*/*Klebsiella* + *Pseudomonas* species. Fourteen of the 18 isolates (14/18, 77%) were CPE and there was 61% (11/18) correlation between the organisms isolated in stool/throat surveillance and later in blood cultures, thus suggesting them to be MBI-LCBI (Table 2). Analysis of pre-transplant surveillance cultures over the last 7 years showed a consistently high prevalence rate of carbapenem-resistant MDRO in stool samples (Table 3).

Of the 64 patients who underwent transplantation, 55 were alive at 100 days (85.9%). One patient died post-autologous transplant (relapsed/refractory HL, cause of death—diffuse alveolar hemorrhage). All the other patients who died were post-allogeneic SCT (seven after haploidentical-SCT and one after matched unrelated donor transplant). There was an association between the risk of all-cause 100-day mortality and a positive stool/throat surveillance culture (all nine patients who died had MDRO in their stool surveillance cultures and six of these nine patients had CPE) (Table 4). However, these data were not statistically significant. Overall, the risk of mortality was

higher in patients with a positive blood culture during transplant than in those without; six out of nine patients who died had a blood culture positive for Gram-negative bacilli (Fisher's exact test, $p = .012$, Table 4). There was no correlation either with the duration of hospital stay or the duration of antibiotic exposure and patient outcomes. None of the patients had central line-associated bloodstream infection (CLABSI), secondary bloodstream infections, or non-CLABSI primary bloodstream infections, as per the current definitions.¹⁵ On analyzing the post-100-day follow-up period until 6 months, another 16 patients died during this period taking the total toll to 25 deaths (25/64, 39.1%). The causes of death were GVHD and infection-related complications, graft failure, and relapse. A similar correlation was seen between all-cause mortality post-100 days of transplant and positive blood culture with Gram-negative MDRO during transplant (Fisher's exact test, $p = .008$, Table 4).

5 | DISCUSSION

In this study, we found a high prevalence of Gram-negative MDRO in the gut flora of patients undergoing HSCTs. They far outnumbered Gram-positive organisms such as VRE and MRSA. There was a strong association between early post-HSCT mortality and documented Gram-negative BSI by these MDROs (Table 4). The rates of Gram-negative bacterial infection are lower in developed countries as compared to developing ones, but the 90–100-day (early) all-cause mortality rates in patients undergoing transplants with a documented BSI are similar and range between 10% and 15%, with high-risk

TABLE 1 Surveillance culture, blood culture (Gram-negative bacterial bloodstream infection), and patient outcome data of 1 calendar year (January 2021–December 2021).

Parameter	Age (years), median $\pm$ SD	Sex	MDRO-positive		MDRO-positive throat surveillance (% of total patients)	CRO prevalence (% of total patients)	Positive blood culture (% of total patients)	100-Day outcome		6-Month post-transplant outcome	
			Male (%)	Female (%)				Alive (%)	Dead (%)	Alive (%)	Dead (%)
Autologous transplant, N = 29	43.4 $\pm$ 15.7	19 (29.7)	10 (15.6)		8 (12.5)	14 (21.9)	6 (9.4)	28 (43.7)	1 (1.6)	21 (32.8)	8 (12.5)
Allogeneic transplant, N = 35	27.2 $\pm$ 24.2	3 (4.7)	2 (3.1)		2 (3.1)	3 (4.7)	2 (3.1)	4 (4.7)	1 (1.6)	2 (3.1)	3 (4.7)
	30.8 $\pm$ 16.02	6 (9.4)	6 (9.4)		3 (4.7)	5 (7.8)	4 (6.3)	12 (17.2)	0	10 (15.6)	2 (3.1)
	25.3 $\pm$ 11.8	14 (21.9)	4 (6.3)		8 (12.5)	8 (12.5)	6 (9.4)	11 (15.6)	7 (12.5)	6 (9.4)	12 (18.7)
Total patients, N = 64	34.7 $\pm$ 17.2	42 (65.6)	22 (34.4)		21 (32.8)	31 (48.5)	18 (28.1)	55 (85.9)	9 (14.0)	39 (60.9)	25 (39.1)

Abbreviations: CRO, carbapenem-resistant organism; Haplo, haploidentical; MDRO, multidrug-resistant organism; MRD, matched related donor; MUD, matched unrelated donor; SD, standard deviation.

diseases and MDR organisms being independent risk factors.¹⁶ Our data also showed a 14.1% (9/64) 100-day all-cause mortality in patients with Gram-negative BSI. The use of pre-transplant stool and throat swab surveillance cultures guided empirical antibiotic policy and subsequent antibiotic escalations. This allowed better management of patients with complicated infections and led to a good early post-transplant outcome, comparable with other transplant centers. Hence, this study emphasizes the need for surveillance cultures prior to SCTs, in areas with high MDRO endemicity and justifies the added expense incurred by the laboratory to perform such tests.

In this report, the pre-transplant surveillance cultures were positive for MDRO in 85.9% of patients, with 48.5% of the isolates being positive for carbapenemase-producing genes; the most common types of CPE MDROs were *E. coli* and *Klebsiella*, which produce NDM/OXA-48 or both. A recent study from India showed a relatively high prevalence (6.1% fecal carriage rate) of CPE in the general population.¹⁷ Similar data have been reported from the HSCT unit at Vellore in South India, where the prevalence of common drug-resistant isolates in fecal surveillance cultures was 57.7% (134 out of 232 patients), with 81% being positive for ESBL and 11.5% for carbapenem resistance.⁴ Of 732 stool surveillance cultures in 242 adult oncology patients (50% HSCT) admitted to the Johns Hopkins Medical Centre in the USA, MDRO was present in 251 cases, the most common being VRE (52.2%); ESBL-producing organisms was seen in 22.2%; and carbapenemase-producing organisms in only 4.4% (10 times lower than in this study).⁵

A global report has put the Indian sub-continent as an epicenter for the emergence of metallo-beta-lactamase-producing CPE, viz. the NDM.¹⁸ In hematologic malignancy patients, this figure is even higher, as we found in this study. One potential source of these MDROs is antibiotic selection pressure. In leukemia patients, induction and consolidation chemotherapies often get complicated by neutropenic sepsis, requiring prolonged use of broad-spectrum antibiotics, leading to a selection of antibiotic-resistant bacteria and alteration in the gut microbiota. Such patients, when posted for HSCT, pose a real challenge with regard to empirical antibiotic policy, antibiotic stewardship, and overall management of the neutropenic phase during the conditioning and pre-engraftment periods. A study from Spain reported that hospitalization for longer than 7 days, respiratory infection, and administration of any antibiotic were associated with increased MDR *K. pneumoniae* colonization (as demonstrated by surveillance cultures from rectal swabs, oro-pharyngeal swabs, and bronchial aspirates). Patients who received antibiotics for more than 48 h were colonized earlier than patients who did not receive antibiotic treatment (hazard ratio: 2.16).¹⁹ Another study from Mongolia reported that Gram-negative MDRO colonization rate doubled during hospitalization.²⁰ In a study from the University of Tübingen in Germany, the effect of antibiotic selection pressure was determined through sequence-based metagenomics. The study found a strong positive selection for class D beta-lactamases and the authors suggested that antibiotic selection for therapy be optimized to ensure lowest selection pressure in terms of antibiotic resistance.²¹ However, such algorithms are yet to be implemented on a large scale in transplant centers and are still at the

TABLE 2 Prevalence of multidrug-resistant organism (MDRO) in stool and throat surveillance cultures, blood cultures, and their correlation.

MDRO detected in surveillance culture	Resistance phenotype/genotype	Stool surveillance ^a (% of total patients), N = 64 patients	Throat swab surveillance ^a (% of total patients), N = 64 patients	Positive blood culture, N = 18 patients ^b	Proportion of patients with same organism in surveillance and blood cultures ^c
<i>Escherichia coli</i>	AmpC	19 (29.68)	2 (3.1)	1	0/1
	ESBL	22 (34.37)	2 (3.1)	3	0/3
	Carbapenemase—NDM, OXA	16 (25)	2 (3.1)	5	4/5
<i>Klebsiella</i> species (<i>pneumoniae</i> , <i>oxytoca</i>)	AmpC	5 (7.8)	4 (6.2)	3	0/3
	ESBL	3 (4.6)	–	–	–
	Carbapenemase—NDM, OXA	15 (23.43)	4 (6.2)	7	6/7
<i>Pseudomonas aeruginosa</i>	Carbapenemase—VIM	–	3 (4.6)	3	1/3
Vancomycin-resistant <i>Enterococci</i>		4 (6.2)	3 (4.6)	–	–
<i>Salmonella enterica</i> (serovar Typhimurium)		1 (1.5)	–	–	–
<i>Burkholderia cepacia</i> (Carbapenem resistant)		–	1 (1.5)	–	–

Abbreviations: AmpC, Ambler class C-producing beta-lactamases; ESBL, extended spectrum beta-lactamase; NDM, New Delhi metallo-beta-lactamase; VIM, Verona integron-encoded metallo- β -lactamase.

^aAll 64 patients underwent stool and throat cultures of which some patients had multiple MDRO growing in cultures.

^bSix patients had multiple isolates from the blood culture.

^cCorrelation: The MDRO, which was isolated from surveillance cultures prior to transplant, was identical to that isolated from blood culture during the post-transplant period.

TABLE 3 Surveillance culture data, multidrug-resistant organism (MDRO) prevalence rates, and antibiotic resistance patterns over the last 7 years.

Parameter	Stool surveillance (% MDRO)	Throat surveillance (% MDRO)	Meropenem resistance—stool surveillance (%)	Meropenem resistance—throat surveillance (%)
2015	87.3%	35.4%	38.8	31.0
2016	94.1%	49.2%	32.7	27.7
2017	90.1%	38%	33.2	37.8
2018	77.2%	33%	41.5	61.8
2019	85.5%	20%	34.8	47.6
2020	60.7%	9.9%	50.0	60.0
2021	73.1%	18.2%	42.8	52.8

TABLE 4 Correlation of surveillance cultures and blood culture positivity rates with 100-day and post-100-day mortality rates.

Parameter		100-Day mortality		100 days–6 months post-transplant	
		Number of deaths (% of all total patients)	p-Value	Number of deaths (% of total patients)	p-Value
MDRO in surveillance cultures (stool and throat)	Positive, N = 55	9 (14)	.58	15 (23.4)	.67
	Negative, N = 9	0		1 (1.5)	
Blood cultures	Positive, N = 18	6 (9)	.012	9 (14)	.008
	Negative, N = 46	3 (4.6)		7 (10.9)	
Total deaths		9 (14)		16 (25)	

Abbreviation: MDRO, multidrug-resistant organisms.

p values are traditionally depicted in italics. The significant values have been shown in bold.

exploratory level. Other sources of MDRO colonization would include the intake of food and water contaminated with resistant bacteria or antibiotic residues. Previous studies have reported antibiotic use in animal husbandry practices and may attribute to the high prevalence of MDRO in the general population.²²

GVHD rates are also higher in patients with MCI-LCBI and this is an established clinical problem affecting overall HSCT outcomes.^{1,23} As per the criteria of the Centers for Disease Control and Prevention, one of the diagnostic criteria of MBI-LCBI is an allogeneic HSCT recipient who develops a grade III or IV gastrointestinal GVHD within the past year and also has a positive blood culture, both documented during the same hospitalization.²⁴ Therefore, it becomes imperative to reduce MBI-LCBI as the outcomes with grade III–IV gut GVHD are extremely dismal.²³

Gram-negative sepsis as a predictor of poor outcome in hematologic patients is an established fact. A study from a tertiary-care hematology institute in the same geographical region showed the incidence of MDR *E. coli* and *Klebsiella* to be as high as 50% and most of them were carbapenem resistant (resistance ranging from 70% to 90%). The authors found that pulmonary infections by Gram-negative bacteria and acute myeloid leukemia were independent predictors of mortality in hematology patients with febrile neutropenia.²⁵ Although allogeneic transplants are performed in highly controlled and sterile environments, the incidence of bloodstream infections in this setting is also quite high and ranges between 10% and 25% in various studies.^{26–28} A recent audit published from another center in India quotes a higher figure of nearly 30%, of which carbapenem-resistant Enterobacterales (CRE) constituted nearly 70% of the isolates.²⁹ Our own data published in 2014 and 2015 showed the prevalence of colistin-resistant *Klebsiella* and high rates of CRE in leukemia patients undergoing intensive chemotherapy.^{12,29} Recent data from the Indian Council of Medical Research's Antimicrobial Resistance Surveillance Network, on the other hand, showed that out of the all the carbapenems, meropenem showed 67% susceptibility in Enterobacterales, and 64.7% of the isolates were sensitive to meropenem among *P. aeruginosa*.³⁰ Although the prevalence of MDR/XDR organisms is high in this geographical region, with regard to our center, the incidence of MDR infections has not shown a significant rise in recent years as evidenced by our recent publication of an analysis of MDR and XDR organisms causing bloodstream infections over the last 11 years. These data were, however, not limited to HSCT recipients only but included patients with hematological and solid organ malignancies. Positive blood culture rates were 12.2%–15.8% overall and MDR/XDR BSI rates per 1000 inpatient encounters were 55%–89.91% and 18.33%–31.11% for 3GCR Gram-negative bacilli and carbapenem-resistant Gram-negative bacilli, respectively.³¹

Efforts to control BSI are of paramount importance if transplant outcomes are to be improved in MDR endemic regions such as ours. Analysis of the surveillance cultures for the preceding 7 years at our institute also highlights the constant and high levels of MDR endemicity in the patients' gut. This is a non-modifiable factor in the present day until further research into the gut microbiome and how it may be modified is done. Out of the modifiable factors, hand hygiene is first

area that needs to be revisited, as it is the first and essential component of standard precautions. Our center showed a CLABSI rate of 1.48 per 1000 catheter days, which is very low and comparable with some of the best transplant centers in the world.³² Other factors, such as air and water quality, surface and environmental sterilization and disinfection, are also important metrics that need frequent audits and quality checks. Genotyping for carbapenemase-producing genes by PCR is now standard³³ in most transplant centers, including ours, and makes it possible to salvage most of the patients with appropriate antibiotics, albeit with increased morbidity. The overall cost of the transplants also increased for the patients with MDR infection. Another way of proving patient-to-patient transmission of MDROs due to poor hand hygiene is to use next-generation sequencing, pulsed field gel electrophoresis, and PCR-restriction fragment-length polymorphism, techniques that determine the clonality of bacterial strains.³⁴ However, we do not have access to the above technologies. Studies on the gut microbiome and the use of fecal microbiota transfer in SCT settings are gaining momentum and are key areas of research that may solve some of the above-mentioned challenges.³⁵

There are some limitations of this audit. Due to a relatively small number of patients and a very high prevalence of MDRO in the study population, a statistically significant correlation between MDRO-positive patients and positive blood cultures could not be established. Since the study was retrospective in nature, we acknowledge the fact that without a sample size calculation, finding a statistically significant correlation between MDRO isolation and clinical outcome is difficult to achieve.

6 | CONCLUSION

The analysis of SCT outcomes at our center shows how MDRO infections affect the former, with higher all-cause mortality rates. With the existing armamentarium of antibiotics, performing allogeneic transplants in patients harboring MDROs is going to be very challenging in the years to come. Our practice of performing pre-transplant stool and throat surveillance cultures offers a heads-up with regard to empirical antibiotic policy and also allows for rapid escalation to appropriate drugs when the patient is sick, bypassing standard guidelines for febrile neutropenia. These practices have helped us maintain favorable transplant outcomes, with overall mortality rates comparable with some of the leading centers in the world. Further research should focus on studying the gut microbiome, the factors affecting it, and the ways of modulating it in these high-risk patients, possibly with the use of fecal microbiota transfer and other novel therapies.

AUTHOR CONTRIBUTIONS

Manuscript drafting, statistical analysis, data collection and collation, concept of study, clinical care, and patient management: Shouriy Ghosh. *Microbiological data collection, statistical analysis, and manuscript editing and concept:* Sanjay Bhattacharya. *Microbiological data collection, analysis, and concept:* Gaurav Goel. *Microbiological data collection and analysis:* Rasika Avinash Deshmukh. *Data collection and analysis:* Rizwan Javed.



Clinical data collection, archival, hand hygiene audit, and infection control audits: Mita Roychowdhury. Statistical analysis: Subir Sinha. Infection control audits and data management: Maitrayee Sarkar De. Clinical management, concept, and data analysis: Arijit Nag. Clinical management: Jeevan Kumar, Saurabh Jayant Bhavne, and Reena Nair. Concept of study, manuscript editing, data analysis, and overall supervision: Mammen Chandy.

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The authors declare no conflicts of interest.

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CONSENT FOR PUBLICATION

Yes.

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SUPPORTING INFORMATION

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